

Optimized Tajik Language Speech Recognition Training Schedule

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1. Objective

To develop a compact and effective speech recognition model for the Tajik language within a 12-day timeline. The model will transcribe spoken Tajik audio into text, using pre-trained architectures like Whisper-small or Wav2Vec2, adapted to work within laptop GPU constraints.

2. Requirements

- **Data Collection:**
 - Gather Tajik speech audio from radio, interviews, podcasts, and public recordings.
 - **Preprocessing:**
 - Clean audio (WAV preferred), segment it, align it with transcripts, normalize file formats.
 - **Model Architecture:**
 - Use a lightweight pretrained model such as Whisper-small or HuggingFace's Wav2Vec2.
 - **Training Environment:**
 - Python 3.9+, PyTorch, Hugging Face Transformers
 - Laptop GPU: NVIDIA RTX 30-60 series (4-8 GB VRAM)
 - Training Time: 1-3 hours/day for subset training
 - Tools: Google Colab (Pro), or local WSL2/Linux with CUDA
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3. Condensed Training Schedule (12 Days)

Day 1-3: Data Collection & Preprocessing

- Gather and segment audio datasets
- Clean, tokenize, and align audio-text pairs

Day 4-5: Model Setup & Configuration

- Select model (Whisper-small, Wav2Vec2)
- Configure environment, verify dataset format

Day 6-8: Training (Subset)

- Train using small batches compatible with limited GPU
- Adjust learning rates and batch sizes as needed

Day 9-10: Evaluation & Fine-tuning

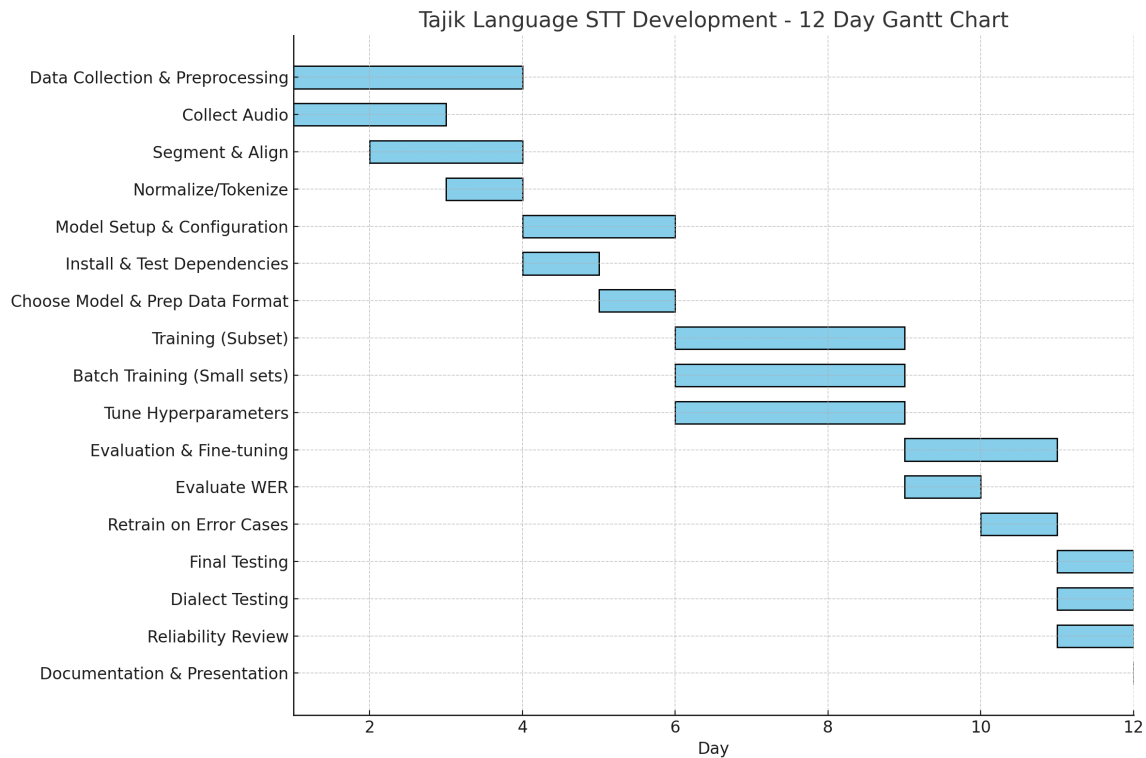
- Measure performance (Word Error Rate)
- Retrain on specific segments to improve accuracy

Day 11: Final Testing

- Test with unseen audio samples
- Note WER and model reliability across dialects

Day 12: Documentation & Presentation

- Summarize tools, data, and results
- Prepare presentation/report for stakeholders



4. Pre-built Model Option

- Use existing multilingual STT models like Whisper or Vosk
 - Faster deployment, avoids full training
 - May need minor fine-tuning for Tajik
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5. Recommendation

This revised 12-day plan provides a realistic, resource-aware schedule for building a Tajik STT module. Choose custom training for control and accuracy, or use pre-trained models for faster results on a limited budget and hardware.
